

Effectiveness of Switching CGRP Monoclonal Antibodies in Non-Responder Patients in the UAE: A Retrospective Study.

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Calcitonin gene-related peptide monoclonal antibodies (CGRP mAbs) have shown promising effectiveness in migraine management compared to other preventative treatment options. Many questions remain regarding switching between antibody classes as a treatment option in patients with migraine headaches. This preliminary retrospective real-world study explored the treatment response of patients who switched between CGRP mAb classes due to lack of efficacy or poor tolerability. A total of 53 patients with migraine headache switched between three of the CGRP mAbs types due to lack of efficacy of the original prescribed CGRP mAbs, specifically eptinezumab, erenumab, and galcanezumab. Fremanezumab was not included due to unavailability in the UAE. Galcanezumab and eptinezumab target the CGRP ligand (CGRP/L), while erenumab targets CGRP receptors (CGRP/R). The analysis of efficacy demonstrated that some improvements were seen in both class switch cohorts (CGRP/R to CGRP/L and CGRP/L to CGRP/R). The safety of switching between CGRP classes was well observed, as any adverse events presented before the class switch did not lead to the discontinuation of treatment following the later switch. The findings of this study suggest that switching between different classes of CGRP mAbs is a potentially safe and clinically viable practice that may have some applications for those experiencing side effects on their current CGRP mAb or those witnessing suboptimal response.

Evaluating the Effectiveness, Tolerability, and Safety of Eptinezumab in High-Frequency and Chronic Migraine in Real World: EMBRACE-The First Italian Multicenter, Prospective, Real-Life Study.

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We conducted a multicenter, prospective study (EMBRACE) evaluating the real-life effectiveness, safety, and tolerability of eptinezumab (100 mg/300 mg)-a monoclonal antibody targeting the calcitonin-gene-related peptide (anti-CGRP mAb)-in high-frequency episodic migraine (HFEM) or chronic migraine (CM). The primary endpoint was the change in monthly migraine days (MMD) for HFEM or monthly headache days (MHD) for CM at weeks 9-12 compared to baseline. The secondary endpoints included changes in monthly analgesic intake (MAI), Numerical Rating Scale (NRS), Headache Impact Test (HIT-6), Migraine Disability Assessment Scale (MIDAS), Migraine Interictal Burden Scale (MIBS-4), and responder rates. The safety analysis involved 44 subjects; the effectiveness analysis included 26 individuals. Eptinezumab was well-tolerated. In CM patients, eptinezumab significantly reduced MHD (-16.1 ± 9.9 , $p < 0.001$), MAI, NRS, HIT-6, MIDAS, and MIBS-4. In HFEM patients, it significantly reduced NRS, HIT-6, MIDAS, and MIBS-4, though reductions in MMD (-3.3 ± 4.5) and MAI were not statistically significant. Overall, $\geq 50\%$ and $\geq 75\%$ response rates were 61.5% and 30.8%, respectively (60% and 30% in non-responders to subcutaneous anti-CGRP mAbs). The clinical change was rated as much or very much improved by 61.0% of the patients. Eptinezumab demonstrated high effectiveness, safety, and tolerability in real-life among hard-to-treat migraine patients with multiple treatment failures, including anti-CGRP mAbs.

Effectiveness and tolerability of eptinezumab in treating patients with migraine

resistant to conventional preventive medications and CGRP (receptor) antibodies: a multicentre retrospective real-world analysis from Germany.

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Eptinezumab is a monoclonal antibody that targets calcitonin gene-related peptide (CGRP mAb) and is used for migraine prophylaxis. Efficacy data are mainly from clinical trials, real-world data are hardly available yet. Reimbursement policy in Germany leads to eptinezumab mainly being used in patients having failed pre-treatment with other CGRP mAb. To date, it is unclear whether eptinezumab is efficacious and well tolerated in this population and how the treatment response differs from patients who are naive to CGRP mAbs. We analysed clinical routine data of 79 patients (episodic migraine (EM): n = 19; chronic migraine (CM): n = 60) from four different centres in Germany. All patients were treated with eptinezumab (100mg). Differences in monthly headache (MHD), migraine (MMD) and acute medication days (AMD) after three months were analysed. The correlation of response with the number of CGRP mAb failures was evaluated. Significance level has been corrected ($\alpha = 0.017$). After three months MHD, MMD and AMD were significantly reduced. In EM, the median reduction for MHD was 4.0 days (IQR: -6.5 to -1.0; $p = 0.001$), for MMD 3.0 days (IQR: -5.5 to -1.5; $p < 0.001$) and for AMD 2.0 days (IQR: -5.0 to -0.5; $p = 0.006$). In CM, median reduction of MHD was 4 days (IQR: -8.0 to 0.0; $p < 0.001$), 3.0 days (IQR: -6.0 to -1.0; $p < 0.001$) for MMD and 1.0 day (IQR: -5.0 to 0.0; $p < 0.001$) for AMD. All patients were resistant to conventional preventive therapies and most to CGRP mAbs. Fourteen patients had never received a CGRP mAb and 65 patients had received at least one mAb without sufficient effectiveness and/or intolerability (one: n = 20, two: n = 28, three: n = 17). There was a significant association between the number of prior therapies and the 30% MHD responder rate (none: 78.6%, one: 45.0%, two: 32.1%, three: 23.5%, $p = 0.010$). Regarding tolerability, 10.4% (8/77) reported mild side effects. The effectiveness of eptinezumab is significantly reduced in patients who have not previously responded to other CGRP mAbs. However, limitations such as the retrospective nature of the analysis, the small sample size and the short treatment period with only the lower dose of eptinezumab must be considered when interpreting the results.

A real-world prospective observational study of eptinezumab in Asian patients with migraine.

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To evaluate the real-world effectiveness of eptinezumab for migraine prevention in Asian patients. Eptinezumab is a monoclonal antibody that targets calcitonin gene-related peptide (CGRP), a potent vasodilator with an important role in migraine pathophysiology. Although there is robust clinical evidence from pivotal Phase 3 placebo-controlled trials of the efficacy of eptinezumab for migraine prevention, there are limited data on the real-world effectiveness of eptinezumab in Asian patient cohorts. This was a non-interventional, prospective, multisite cohort study of adults with migraine (International Classification of Headache Disorders, 3rd edition criteria) in Singapore who were prescribed eptinezumab (100 mg at baseline and Month 3, administered intravenously) and were followed until Month 6. The primary endpoint was change from baseline in monthly migraine days (MMDs) at Month 3 and Month 6. Secondary endpoints were $\geq 30\%$ and $\geq 50\%$ responder rates, and change from baseline in the Headache Impact Test-6 (HIT-6), Migraine Disability Assessment (MIDAS), Migraine-Specific Quality of Life (MSQ), patient-identified most bothersome symptom (PI-MBS), acute medication use at Month 3 and Month 6, and safety. Enrolled patients (completed = 29/30) had on average 3.4 (SD 2.9) previous preventive

treatments; 29/30 of the patients had trialed at least one previous preventive treatment without benefit. Most had previously trialed oral preventives (87%, 26/30) and anti-CGRP (70%, 21/30). Relative to baseline, mean MMDs decreased by 4.3 days (95% CI 2.1-6.4; $p < 0.001$) at Month 3 and 4.9 days (95% CI 2.1-7.7; $p < 0.001$) at Month 6. At Month 3 and Month 6, 14/30 (47%) and 15/29 (52%) of the patients were $\geq 30\%$ responders, and 6/30 (20%) and 8/29 (28%) patients were $\geq 50\%$ responders, respectively. The number of patients with severe life impairment based on the HIT-6 score (total score 60-78) decreased from 24/30 (80%) at baseline to 19/30 (63%) at Month 3 and 19/29 (66%) at Month 6. The mean MIDAS score decreased by 24.6 points (95% CI 2.82-46.38; $p = 0.028$) at Month 6, and the mean MSQ score increased by 12.2 points (95% CI 5.18-19.20; $p = 0.001$) at Month 3 and 13.6 points (95% CI 4.58-22.66; $p = 0.004$) at Month 6. Most patients reported improvement in the PI-MBS at Month 3 (73%, 22/30) and Month 6 (55%, 16/29). Acute medication use for headache relief decreased by 3.3 days/month (95% CI 1.0-5.6; $p = 0.007$) at Month 3 and 4.7 days/month (95% CI 1.7-7.7; $p = 0.003$) at Month 6. Treatment-emergent adverse events (TEAEs) were reported in 16/30 (54%) patients, mostly mild/moderate in severity. No serious TEAEs led to treatment discontinuation. Quarterly eptinezumab administration was effective and well-tolerated in Asian patients with chronic migraine.

Preventive Treatment of Migraine.

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This article describes strategies for the preventive treatment of migraine including the emerging role of calcitonin gene-related peptide (CGRP)-targeted therapies and introduces novel paradigms for the preventive treatment of migraine. Multiple migraine medications targeting CGRP have been introduced since 2018, including injectable monoclonal antibodies (ie, eptinezumab, erenumab, fremanezumab, and galcanezumab) and oral small-molecule CGRP receptor antagonists (ie, ubrogepant, rimegepant, atogepant, and zavegepant). With the exceptions of ubrogepant and zavegepant, which are approved only as acute treatments, all of these agents have demonstrated efficacy in the preventive treatment of migraine; the monoclonal antibodies and atogepant have evidence of effectiveness in adults with either episodic or chronic migraine. The safety and tolerability profiles of CGRP-targeted therapies in migraine are favorable. The goals of preventive migraine therapy include reducing the frequency, severity, duration, and disability associated with attacks, reducing the need for acute treatment and the risk of medication overuse, enhancing self-efficacy and health-related quality of life, and reducing headache-related distress and interictal burden. Six drugs targeting CGRP (four monoclonal antibodies and two gepants) are now available for the preventive treatment of episodic migraine in adults. The efficacy of CGRP-targeted medications in the acute and preventive treatment of migraine, together with good safety and tolerability, has led to the emergence of new approaches to preventive treatment.

Evaluation of outcomes of calcitonin gene-related peptide (CGRP)-targeting therapies for acute and preventive migraine treatment based on patient sex.

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Women show increased prevalence and severity of migraine compared to men. Whether small molecule calcitonin gene-related peptide receptor (CGRP-R) antagonists (i.e., gepants) and monoclonal antibodies targeting either the CGRP-R or the CGRP peptide might show sexually dimorphic outcomes for acute and

preventive therapy has not been established. We conducted a subpopulation analysis of available published data from FDA reviews to evaluate potential sex differences in the response rates of ubrogepant, rimegepant and zavegepant for acute migraine therapy. Available data from FDA reviews of erenumab, fremanezumab, galcanezumab and eptinezumab, approved CGRP-R and CGRP monoclonal antibodies and of atogepant were examined for prevention outcomes based on patient sex. Preventive outcomes were analyzed separately for patients with episodic migraine and chronic migraine. In women, the three approved gepants produced statistically significant drug effects regardless of dose tested on the FDA mandated co-primary endpoints, the proportion of patients achieving two-hour pain-freedom and the proportion of patients free of their most bothersome symptom at two hours post-dose. In women, the average placebo-subtracted two-hour pain-freedom proportion was 9.5% (CI: 7.4 to 11.6) and the average numbers needed to treat was 11. The free from most bothersome symptom at two hours outcomes were also significant in women. The gepant drugs did not reach statistically significant effects on the two-hour pain-freedom endpoint in the men, with an average drug effect of 2.8% (CI: -2.5 to 8.2) and an average number needed to treat of 36. For freedom from most bothersome symptom at two hours post-dose endpoint, differences were not significant in male patients. The treatment effect in each of the gepant studies was always numerically greater in women than in men. In evaluation of prevention outcomes with the antibodies or atogepant using the change from the specified primary endpoint (e.g., monthly migraine days), the observed treatment effect for episodic migraine patients almost always favored drug over placebo in both women and men. For chronic migraine patients the treatment effects of antibodies were similar in men and women and always favored the drug treated group. Conclusion/Interpretation: Small molecule CGRP-R antagonists are effective in acute migraine therapy in women but available data do not demonstrate effectiveness in men. CGRP-targeting therapies are effective for migraine prevention in both male and female episodic migraine patients but possible sex differences remain uncertain. In male and female chronic migraine patients, CGRP/CGRP-R antibodies were similarly effective. The data highlight possible differential effects of CGRP targeted therapies in different patient populations and the need for increased understanding of CGRP neurobiology in men and women.

Calcitonin gene-related peptide-targeting therapies are a first-line option for the prevention of migraine: An American Headache Society position statement update.

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To provide a position statement update from The American Headache Society specifically regarding therapies targeting calcitonin gene-related peptide (CGRP) for the prevention of migraine. All migraine preventive therapies previously considered to be first-line treatments were developed for other indications and adopted later for migraine. Adherence to these therapies is often poor due to issues with efficacy and tolerability. Multiple new migraine-specific therapies have been developed based on a broad foundation of pre-clinical and clinical evidence showing that CGRP plays a key role in the pathogenesis of migraine. These CGRP-targeting therapies have had a transformational impact on the management of migraine but are still not widely considered to be first-line approaches. Evidence regarding migraine preventive therapies including primary and secondary endpoints from randomized placebo-controlled clinical trials, post hoc analyses and open-label extensions of these trials, and prospective and retrospective observational studies were collected from a variety of sources including PubMed, Google Scholar, and ClinicalTrials.gov. The results and conclusions based upon these results were reviewed and discussed by the Board of Directors of The American Headache Society to confirm consistency with clinical experience and to achieve consensus. The evidence for the efficacy, tolerability, and safety of

CGRP-targeting migraine preventive therapies (the monoclonal antibodies: erenumab, fremanezumab, galcanezumab, and eptinezumab, and the gepants: rimegepant and atogepant) is substantial, and vastly exceeds that for any other preventive treatment approach. The evidence remains consistent across different individual CGRP-targeting treatments and is corroborated by extensive "real-world" clinical experience. The data indicates that the efficacy and tolerability of CGRP-targeting therapies are equal to or greater than those of previous first-line therapies and that serious adverse events associated with CGRP-targeting therapies are rare. The CGRP-targeting therapies should be considered as a first-line approach for migraine prevention along with previous first-line treatments without a requirement for prior failure of other classes of migraine preventive treatment.

Real-world effectiveness of fremanezumab in patients with migraine switching from another mAb targeting the CGRP pathway.

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This is a summary of the research article entitled "Real-world effectiveness of fremanezumab in patients with migraine switching from another mAb targeting the CGRP pathway: A subgroup analysis of the Finesse Study". The discovery of calcitonin gene-related peptide (CGRP) as a therapeutic target in migraine has been one of the greatest achievements in neurology in recent years. Specific antibodies against CGRP bind to it either via a receptor (erenumab) or ligand (fremanezumab, galcanezumab, eptinezumab). Monoclonal antibodies (mAbs) are effective, safe and welltolerated drugs that have been approved for prophylactic treatment if there are at least 4 days with migraine per month. However, in clinical practice, the failure of treatment with mAbs has been observed, and thus the question arises whether it is worthwhile to include treatment using an antibody with a different mechanism of action. The Finesse Study was designed to evaluate the efficacy of fremanezumab in patients with a history of prior treatment failure with other mAbs against the CGRP pathway. Among the 153 patients with priorly failed mAbs, switching to fremanezumab led to a $\geq 50\%$ reduction in the number of days with migraine per month in 42.8% of patients. The conclusion emphasizes that switching to another antibody should be considered in patients with prior therapy failure.

Eptinezumab: First Approval.

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Eptinezumab-jjmr (referred to as eptinezumab hereafter; Vyepti™) is a humanised monoclonal antibody that binds to calcitonin gene-related peptide (CGRP) and blocks its binding to the receptor. CGRP is believed to play a major role in the pathophysiology of migraine. Eptinezumab, delivered by intravenous (IV) administration, is being developed by Lundbeck Seattle BioPharmaceuticals for the prevention of migraine. In February 2020, eptinezumab was approved in the USA for the preventive treatment of migraine in adults. This article summarizes the milestones in the development of eptinezumab leading to this first approval.

Eptinezumab for the treatment of migraine.

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Eptinezumab (ALD-403) is a genetically engineered humanized IgG1kappa directed towards calcitonin gene-related peptide (CGRP), currently in late-stage clinical development. Eptinezumab targets the pathway of CGRP, importantly implicated in migraine pathophysiology, and may represent the first-to-market infusion therapy for migraine prevention. The background for its approval consists in preclinical data and clinical trials. Here, we provide a comprehensive review of molecular pharmacology, pharmacokinetics, metabolism, efficacy and safety investigated in the preclinical and clinical studies, with insight on possible future directions.
